sensing applications. Quantum devices such as single-photon detectors, superconducting qubits, and trapped-ion systems are central to quantum technologies. These materials and devices drive innovation in high-performance computing, energy-efficient electronics, and next-generation sensors, making them a cornerstone of the quantum revolution.

Quantum 2.0 and India

As India emerges as a global leader and one of the world's major economies, investing in quantum technologies is essential to strengthen national security, drive innovation, enhance cybersecurity, and accelerate economic growth—ensuring a competitive edge in a rapidly evolving technological landscape.

India has consistently been at the forefront of quantum science and technology. In 2018, the Quantum Enabled Science and Technology (QuEST) programme was launched with a ₹2.5 billion budget, supporting 51 national quantum laboratories and establishing critical infrastructure for quantum research.

Building on QuEST's success, India launched the I-HUB Quantum Technology Foundation (I-HUB QTF) in Pune in 2020 under the National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS), with a ₹1.7 billion budget dedicated to Quantum Technology (QT) development. I-HUB QTF aims to position India as a global leader in quantum innovation and promote grassroots-level impact through QT-based applications.

The National Quantum Mission (NQM), approved by the Cabinet in April 2023, has a ₹60.03 billion budget spanning 2023-24 to 2030-31. It seeks to accelerate scientific and industrial R&D, foster innovation, and drive QT-led economic growth—solidifying India's role in Quantum Technologies and Applications (QTA).

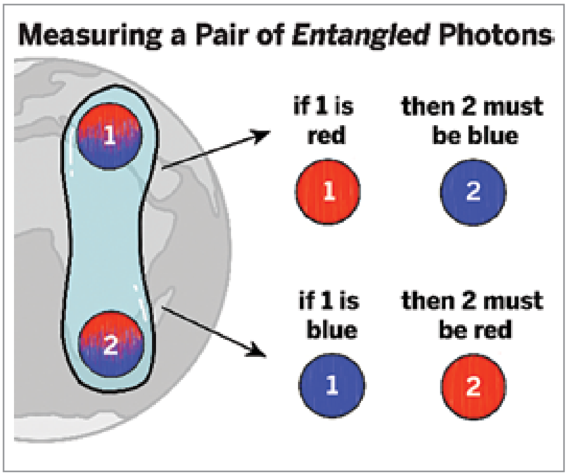

Figure 3: Quantum entanglement (Source: *https://quantumatlas.umd.edu*)

As part of the mission, four thematic hubs (T-Hubs) have been established at leading academic and R&D institutions, focusing on quantum computing, quantum communication, quantum sensing and metrology, and quantum materials and devices. These hubs will advance both fundamental and applied research while encouraging targeted R&D within their respective domains.

The NQM aims to develop quantum computers of up to 1000 qubits using superconducting and photonic technologies in multiple stages. In quantum communication, it targets satellite-based secure links over 2000km, intercity QKD over 2000km via fibre-based trusted nodes, and multi-node quantum networks using quantum memories and repeaters. In quantum sensing and metrology, the mission aims to develop ultra-sensitive magnetometers, high-precision gravity sensors, and atomic clocks for navigation, communications, and geophysical applications. Additionally, the NQM will support innovation in quantum materials, advancing superconductors, novel semiconductors, and topological materials to power quantum computing, secure communication, and high-precision sensing technologies.

In December 2024, the Department of Science and Technology (DST), in collaboration with the All India Council for Technical Education (AICTE), announced a dedicated undergraduate curriculum to create a thriving quantum-trained ecosystem in India as part of the NQM.

As the world moves deeper into the era of Quantum 2.0, the transformative potential of quantum technologies is increasingly evident. From revolutionising computation and communication to enhancing precision in sensing and materials science, quantum advancements are poised to redefine industries and national security frameworks. With strategic investments, a robust research ecosystem, and government-led initiatives like the NQM, India is positioned to become a key player in the global quantum landscape. However, challenges remain, especially in technological scalability and addressing the shortage of skilled professionals. Overcoming these hurdles will require sustained collaboration between academia, industry, and policymakers. As quantum technologies mature, their integration into real-world applications will unlock new possibilities, ensuring a more secure, efficient, and technologically advanced future. END

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